

The Impact of the Metal Interface on the Stability and Quality of a Therapeutic Fusion Protein

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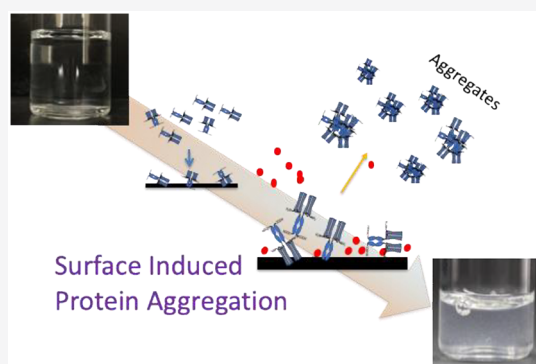
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ABSTRACT: Subvisible particle formation, which occurs after the sterile filtration step of the fill/finish process, is a challenge that may occur during the development of biotherapeutics with complex molecular structures. Here, we show that a stainless steel pump head from a rotary piston pump produces more protein aggregates, past the limit of the acceptable quality range for subvisible particle counts, in comparison to a ceramic pump head. The quartz crystal microbalance was used to quantify the primary layer, proteins irreversibly adsorbed at the solid–liquid interface, and the secondary diffuse gel-like layer interacting on top of the primary layer. The results showed that the mass of protein irreversibly adsorbed onto stainless steel sensors is greater than on an aluminum oxide surface (ceramic pump mimic). This suggests that the amount of adsorbed protein plays a role in surface-induced protein aggregation at the solid–liquid interface.

KEYWORDS: fusion protein, protein adsorption, subvisible particles, solid–liquid interface, protein aggregation



INTRODUCTION

The drug development pipelines for biopharmaceutical companies are increasing, including novel large molecule modalities such as fusion proteins, bispecific proteins, and antibody–drug conjugates.^{1,2} Fusion proteins are a result of two different genes fused together to produce a protein encoded with molecular functionalities not synthesized naturally.³ This may result in a macromolecule with increased size and complexity, which often leads to additional physical and/or chemical instabilities compared to its native parent protein(s). As a consequence, fusion proteins are often more susceptible to non-native irreversible protein aggregation, hereafter referred to as protein aggregation, during manufacturing, storage, and administration.^{2,4}

Denatured proteins or proteins that maintain a structurally altered conformation are more prone to protein aggregation.^{5,6} The aggregated proteins can form a distribution of particle ensembles from a nanometer to micrometer size.⁷ Of particular concern are subvisible particles (SVPs) with dimensions approximately between 1 and 100 μm because of the increased scrutiny from regulatory agencies such as the U.S. Food and Drug Administration.^{8–18} These particles may increase the risk of immunogenicity relative to the same monomeric protein, as demonstrated by various *in vitro* and animal models.^{14–16} Whether SVP-mediated immunogenicity occurs or not seems to depend on the size and concentration of aggregates, the stress condition used to generate the aggregates, the protein itself, and the model used in the study (e.g., *in vitro* or mouse-

based models). The relationship between these studies and immunogenicity in the clinic remains poorly understood. It is also critical to note that the amount of aggregates/SVPs used in such studies is typically much higher than what is normally found in commercial biotherapeutic products, which are expected to comply with limits set in the United States and European Pharmacopeias.^{17,18} Currently, there are safe, marketed products that were approved prior to the routine monitoring and control of SVPs. Nevertheless, there is a need to better understand the subvisible particle (SVP) formation to more effectively identify and disrupt potential particle formation pathways to ensure product quality, safety, and regulatory expectations.⁷

A number of factors can cause protein aggregation and SVP formation, which can occur in bulk solution^{19,20} or at solution interfaces (e.g., air–liquid^{21–24} solid–liquid^{25–29}). Interface-mediated aggregation is of interest because protein solutions are exposed to different surfaces throughout its manufacturing and product usage.³⁰ Contact with an incompatible interface during manufacturing, transportation, storage, or administration may facilitate SVP and/or visible particle (VP) formation. For example, monoclonal antibody (mAb) solutions formed

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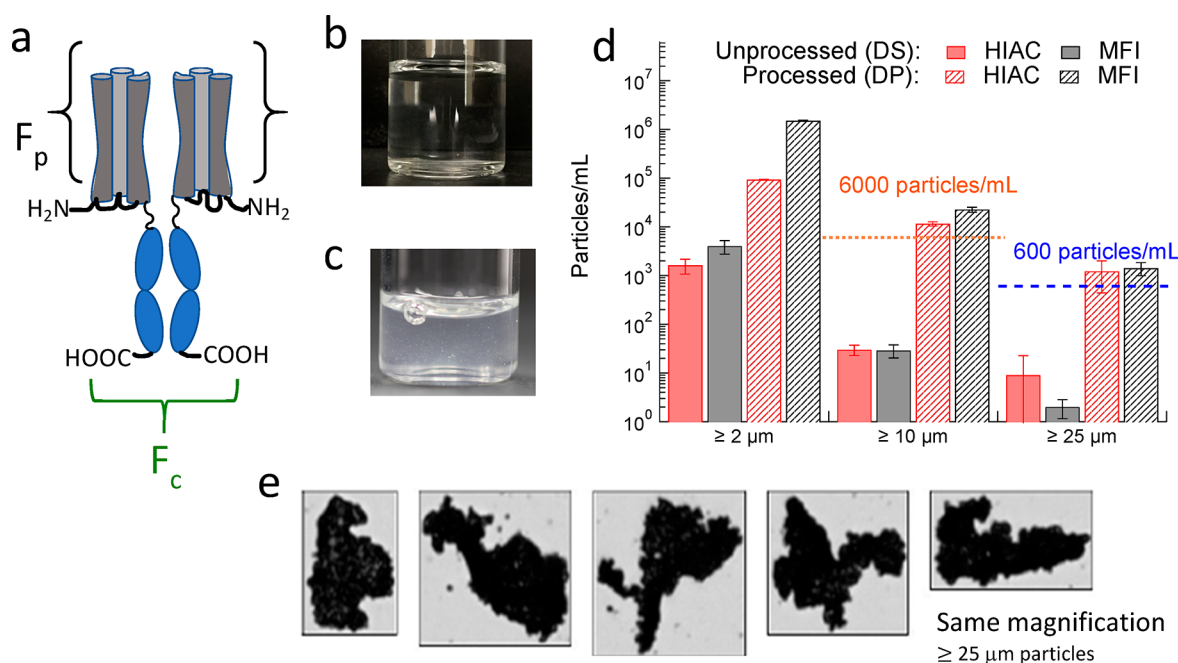


Figure 1. (a) A schematic of the fusion protein, F_cF_p , illustrates that the scaffold is similar to that of mAb. Visual inspection of (b) the drug substance (DS), a visually clear protein solution that has not been processed through the SS pump, and (c) the drug product (DP), an opalescent protein solution as a result of processing through the SS pump. (d) Particle size count measured by light obscuration (HIAC) and flow imaging (FI) for drug substance (solid shading) and drug product (hatched). The concentration of the solution is 50 mg/mL. The horizontal dashed lines indicate the threshold for acceptable particle counts defined by USP (787) for each size category. (e) Images of SVPs as captured by FI. Particle size count uncertainties, ± 1 st. dev., reported for HIAC and FI are from an aggregated 30 mg/mL fusion protein solution, $n = 5$.

SVPs when exposed to shear in the presence of stainless steel^{25,27,31} or silica surfaces.²⁶ At the air–water interface, changes in interfacial area and subsequent thin film rupture can also promote protein aggregation.^{22,32} In previous work, SVP formation of the National Institute of Standards and Technology (NIST) mAb was demonstrated to correlate with protein adsorption, layer formation, and conformational change at the stainless steel–solution interface.²⁵ Given that interfaces and surfaces could play a role in SVP/VP formation, investigations of protein adsorption behavior may provide insight into surface-induced protein aggregation. Protein adsorption has been extensively investigated and reviewed in the scientific literature,^{26,31,33–42} but little is known and/or understood about how/if this adsorption behavior can relate to SVP/VP formation.

In this work, we determined that pump stresses generated from using a stainless steel (SS) rotary piston pump are the root cause for increased SVP formation for a fusion protein. Upon a single passage through the SS rotary pump, this protein demonstrated substantial particle formation, impacting product quality. Interestingly, the fusion protein solution (Drug Substance, DS) did not exhibit substantial quality issues until it was filled into vials (Drug Product, DP) using the SS rotary piston pump. The quartz crystal microbalance with dissipation monitoring, a surface-sensitive analytical technique, was used to quantify the differences of protein surface interactions between two surfaces, SS and aluminum oxide (Al_2O_3 , ceramic mimic). We replicated the exposure of these solid surfaces to dilute and concentrated protein solutions to investigate the film formation process experienced at these interfaces during production. It was observed that the SS surface adsorbed more fusion protein compared to Al_2O_3 . We discuss these results in the context of surface-mediated SVP formation.

EXPERIMENTAL PROCEDURES

The use of commercial products and instrumentation within this publication is for information only; it does not imply recommendation or endorsement by the National Institute of Standards and Technology.

Materials and Methods. Materials. The fusion protein (see schematic Figure 1a) was produced and supplied by AstraZeneca. The fusion protein is composed of a three-repeat construct of an extracellular domain derived from a tumor necrosis factor superfamily protein linked through a G4S repeat sequence to a humanized IgG4p Fc domain. The dimerization of the Fc domain generates a dimer of trimer structure. The full active molecule molecular weight of the nonreduced and deglycosylated form was 148 005 Da by mass spectrometry. Size exclusion chromatography coupled with multi-angle light scattering was measured to be 159 000 Da. The fusion protein was formulated in sodium phosphate buffer solution (pH = 6.9) containing arginine hydrochloride in deionized water. Sodium phosphate monobasic monohydrate, sodium phosphate dibasic heptahydrate, polysorbate 80, water for injection, and L-arginine hydrochloride were all used and purchased from Avantor or J.T. Baker. All materials were at least United States Pharmacopeia grade.

Protein Preparation. Fusion protein formulations were prepared using ultrafiltration and diafiltration (UF/DF) using a 30 K molecular mass cutoff membrane. The diafiltration was performed at 50 mg/mL with 6 diafiltration volumes for complete buffer exchange. The fusion protein was concentrated to >110 mg/mL and recovered. The fusion protein was then diluted to the various concentrations with the diafiltration buffer. In experiments where pumping was performed, polysorbate 80 was added for a final surfactant concentration of 0.02% mass to volume fraction. The final solutions were

filtered through a 0.2 μm PVDF filter prior to the experiments. The solution concentrations were confirmed using UV spectroscopy measured at wavelength 280 nm. The purified solutions were stored in polyethylene terephthalate glycol (PETG) Nalgene bottles at 4 $^{\circ}\text{C}$.

Protein Characterization. The target concentration was verified using ultraviolet light (DropSense 96 polychromatic microplate reader) and pH measured with a Mettler Toledo meter (SevenExcellence). The dynamic viscosity of the fusion protein solution was measured on a cone and plate rheometer at 10 000 s^{-1} (Anton-Parr). Size exclusion chromatography (Agilent 1200 HPLC) and nonreducing gel electrophoresis were used to characterize protein purity. This methodology and data are shown in the [Supporting Information](#). Capillary isoelectric focusing measured the protein's isoelectric point ranges from $\text{pH} = 7.1$ to 8.6.

Rotary Piston Pump. The stainless steel piston rod and casing (Bausch & Strobel Ilshofen, Germany) and a ceramic piston rod and casing were paired with a Colanar FSR1000 pump. Inlet tubing was connected to the container holding purified protein to be used as the bulk reservoir. The formulated solution was pumped through the rotary piston pump one time. The pumped solution was collected and stored in a PETG Nalgene container. Prior to collection, the pump was primed with a solution to remove any visible air bubbles in the inlet and outlet tubing. The gap of the stainless steel and ceramic piston was $90\ \mu\text{m} \pm 10$ and $10\ \mu\text{m} \pm 0\ \mu\text{m}$, respectively, as measured by a digital caliper (Digimatic) with 10 μm uncertainty.

Peristaltic Pump. The peristaltic pumping was performed using a Flexicon FP6 (Denmark). The tubing was connected to a container holding the protein solution and passed through the rotary mechanism attached to a stainless steel filling nozzle. The formulated solution was pumped through the system one time and collected into PETG Nalgene containers. Prior to collection, the tubing was primed with a solution to remove any visible air bubbles in the inlet and outlet tubing. The velocity of the pump was set to match the same filling velocity of the rotary piston pump (≈ 60).

Quartz Crystal Microbalance. Adsorption of protein solutions on SS (US316) and Al_2O_3 was quantified using the quartz crystal microbalance (QCM-D, Q-Sense AB, Sweden). The sensors were purchased from Nanoscience Instruments. A detailed description of the instrumentation and measurement technique is described elsewhere.^{43,44} Prior to use, sensors were cleaned with a protocol to ensure measurement reproducibility. The SS sensors were rinsed with ethanol, deionized water, and dried with N_2 gas, fresh from the box as delivered. After the adsorption experiment, the stainless steel sensors were cleaned with a 2% mass fraction Hellmanex III solution. The cleaning process in detergent was monitored by QCM-D following the desorption of adsorbed species. After reaching the steady state during the cleaning process, the solutions were rinsed with deionized water, ethanol, an deionized water and dried with N_2 gas. Prior to use for the next adsorption experiment, the sensor was placed under UV Ozone (UVO Cleaner Jelight Company Inc.) for 15 min. For Al_2O_3 , prior to the first experiment, the sensors were treated with UV Ozone for 15 min and rinsed with ethanol and deionized water from the box as delivered. All other procedures were the same as with SS sensors.

QCM-D experiments were performed as a series of sequential steps of exposure and wash: (1) a baseline was

established with respect to adsorbate species of interest to be measured. For example, to monitor the adsorption of protein species, the buffer solution was used as a reference baseline. If the components of the buffer were of interest to monitor, water was used as the reference baseline solution. (2) After a baseline was established, the sample solution was introduced into the flow cell at a flow rate of 200 $\mu\text{L}/\text{min}$ for 5 min. The volume of the sensor chamber is 62 μL . The flow was then turned off until frequency and dissipation values reached steady state. (3) The solution was then rinsed at 200 $\mu\text{L}/\text{min}$ with the baseline solution until the frequency and dissipation values reached steady state. With protein, steps, (2) exposure and (3) wash, were repeated for sequential additions of protein concentrations [0.1, 1.0, 10.0, 25.0, 50.0, 110.0] g/mL.

The QCM-D data was analyzed in general with a simple two-layer model consisting of the primary and secondary layer with each layer having physical properties of mass (m), viscosity (η), modulus (G), and density (ρ).²⁵ The primary layer is rigid, whereas the secondary layer is viscoelastic. The primary layer is the closest to the surface, while the secondary layer is located on top of the primary layer. The multilayer is in contact with a Newtonian fluid environment, where the viscosity was determined by rheometry. More information about model fitting is discussed in the [Supporting Information](#). In some circumstances, this model has simpler limits. For example, when the harmonic overtones for both frequency and dissipation have minimal separation ($< 2\ \text{Hz}$), and the dissipation is $\ll 10^{-5}$, the model converges to the Sauerbrey limit⁴⁵ of a rigid film adsorbed onto the quartz resonator, in the presence of the reference baseline solution. QCM-D data with these features were analyzed using the Sauerbrey equation to calculate the mass adsorbed. The adsorbed mass, to construct an adsorption isotherm, was reported from the data obtained during the wash periods.

Particle Size Measurements. Particle size distribution of micrometer-sized particles was measured in two different ways. One method used a DPA 5200 Microflow Imaging (MFI) system (Protein Simple, San Jose, California) to characterize SVPs from the 2 to 100 μm equivalent circular diameter (ECD) in size produced from the interaction of formulations with the two different pump heads. The flow cell was flushed with water and alkaline detergent in between each measurement to ensure a clean flow cell. During the measurement, the FI flow cell was first purged with 200 $\mu\text{L}/\text{min}$ of the sample, and 600 $\mu\text{L}/\text{min}$ of the sample was used for analysis. Light obscuration and the HIAC 9703+ liquid particle counter (Beckman Coulter, Indianapolis, IN) was the other method used to measure SVPs. The instrument was flushed four times, 1 mL each, with clean deionized (DI) water.

RESULTS

Upon filling the fusion protein into vials, a noticeable change in the turbidity of the solution was observed, as shown in [Figures 1b,c](#). [Figure 1d](#) shows the particle counts exceeded thresholds set by USP <787> after the filling process. Particle size counting techniques for SVP were consistent in measuring unacceptable particle size counts. [Figures 1e](#) shows images of the particles identified by flow imaging (FI). The irregular shapes and size distribution suggest that the particles are composed of protein as opposed to extrinsic particles or intrinsic particles from the vial. This was confirmed by vibrational spectroscopy and scanning electron microscopy. These measurements did not detect any metallic or foreign

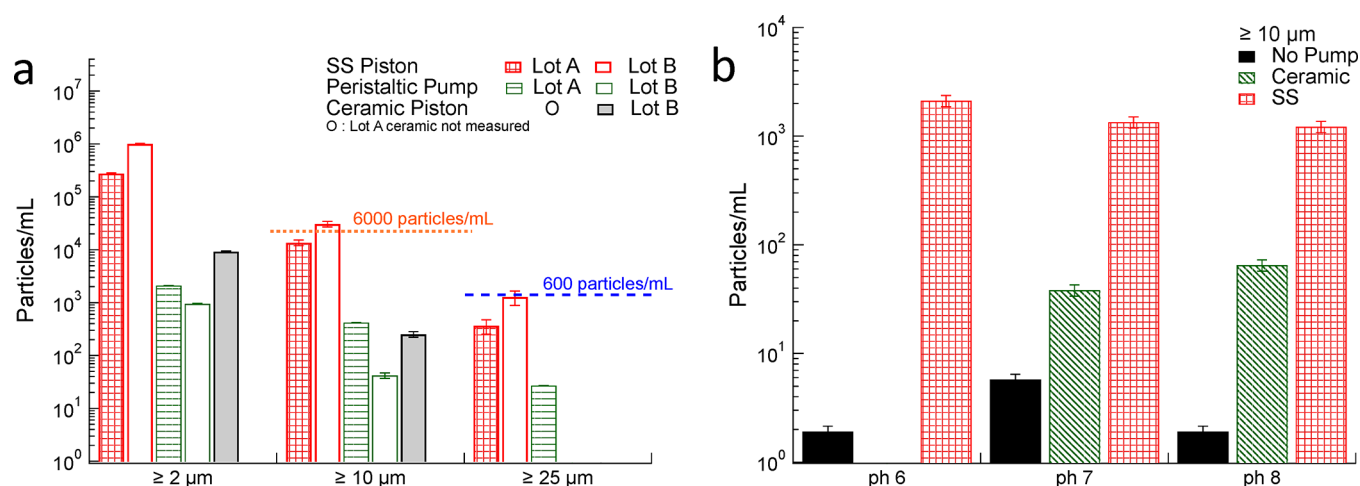


Figure 2. SVP counts measured using FI for the (a) drug product filled using different pumps and (b) SVP $\geq 10 \mu\text{m}$ in the drug product formulated at different pH conditions and filled with different pumps. SVP counts for pH 6 pumped with a ceramic rotary pump were not measured. The concentration of the solution is 50 mg/mL. Particle size count uncertainties, ± 1 st. dev., reported for HIAC and FI are from an aggregated 30 mg/mL fusion protein solution, $n = 5$.

signatures (see [Supporting Information](#)). Interestingly, no SVP issues were observed prior to filling the formulated fusion protein (drug substance/DS) into vials (drug product/DP). DS refers to a protein formulation that has not been processed, and DP refers to a formulation that has been subjected to the fill/finish process. The change in visual appearance between drug substance and drug product suggests manufacturing conditions during processing led to protein destabilization.

Two large scale lots of the partially purified drug substance were investigated to understand potential sources of variability in the purification processes, which could contribute to increased numbers of SVPs observed immediately after vial filling. Factors under consideration were arginine viral inactivation hold times (v_{inac}) and freeze–thaw cycles rates ($F/T = \text{slow/fast}$). The processing conditions for the two lots were ($v_{\text{inac}} = 24$ h, $F/T = \text{slow}$) and ($v_{\text{inac}} = 0$, $F/T = \text{fast}$), respectively, as these were determined to be the worst-case boundary conditions. A peristaltic or rotary piston pump, constructed from SS or ceramic, was used to determine the impact of pump type/contact material. [Figure 2a](#) shows the particle size counts under various processing conditions for the two different lots and the different pumping conditions. The fusion protein did not form appreciable amounts of subvisible particles when using the peristaltic or rotary piston pump composed of ceramic. From these two pumping conditions, the results demonstrate that downstream purification processes did not contribute to the root cause for the sensitivity of the fusion protein to form particles during DP filling. Using the SS rotary piston pump, however, caused SVP formation regardless of the DS lot's processing history.

To further investigate the impact of pumping on SVP formation, the pH of the fusion protein formulation was changed and tested using the two different piston pump head materials, ceramic and SS, shown in [Figure 2b](#). The results show the SS pump head is further implicated in the increased amount of submicrometer particles. Other formulation conditions were also tested to determine the impact of formulation on SVP formation mediated by surfaces. These other results are outside of the intended scope of this manuscript and will be reported in a follow-up study.

To understand how this formulated protein interacts at the solid–liquid interface, we used the quartz crystal microbalance with dissipation monitoring (QCM-D) to assess the adsorption behavior of adsorbates such as excipients, salts, and protein, in formulation. QCM-D is a surface-sensitive analytical technique that monitors changes in mass due to a frequency shift of the quartz crystal resonator. Using this approach, we sought to understand the adsorption behavior at the two different surfaces in the cases of concentrated and dilute protein environments. During manufacturing, surfaces are exposed to protein/excipient concentrations, potentially causing the protein, excipients, or both to adsorb onto these surfaces. This adsorbed film is subject to various physical stresses during fill/finish such as flow and perhaps “mechanical scraping” (rotary piston mechanism). To mimic the adsorption conditions, the QCM-D sensors were exposed to sequential adsorption (protein solution) and washing (buffer) periods. An adsorption period is defined when the solution plus adsorbate species of interest flows over the QCM-D sensor. The wash period occurs after the adsorption period and when buffer alone is washed over the sensor. Sequential changes in an increasing concentration from 0.1, 1.0, 10.0, 25.0, 50.0, to 110.0 mg/mL investigate the sensitivity of adsorption to concentration (analogous to an adsorption isotherm) and elaborate the evolution of the adsorption process. When analyzed, these experimental results describe various adsorption phenomena that occur during manufacturing. Specifically, they determine how much is adsorbed and whether such adsorption is reversible or irreversible.

Monitoring the different periods of adsorption followed by washing tests how strongly an analyte species is bound to a surface. Reversibly bound species adsorb and desorb off the surface back into solution during a washing period. Species that remain after this washing period are irreversibly adsorbed onto the surface. This material attached directly to the surface is called here the primary layer, whereas any beyond is referred to as the secondary layer. The primary layer irreversibly adsorbs on the solid surface, and the secondary layer is reversibly equilibrated on top of the primary layer. Using an appropriate model, the QCM-D data can quantify this strength of adsorption through changes of adsorbed mass. The

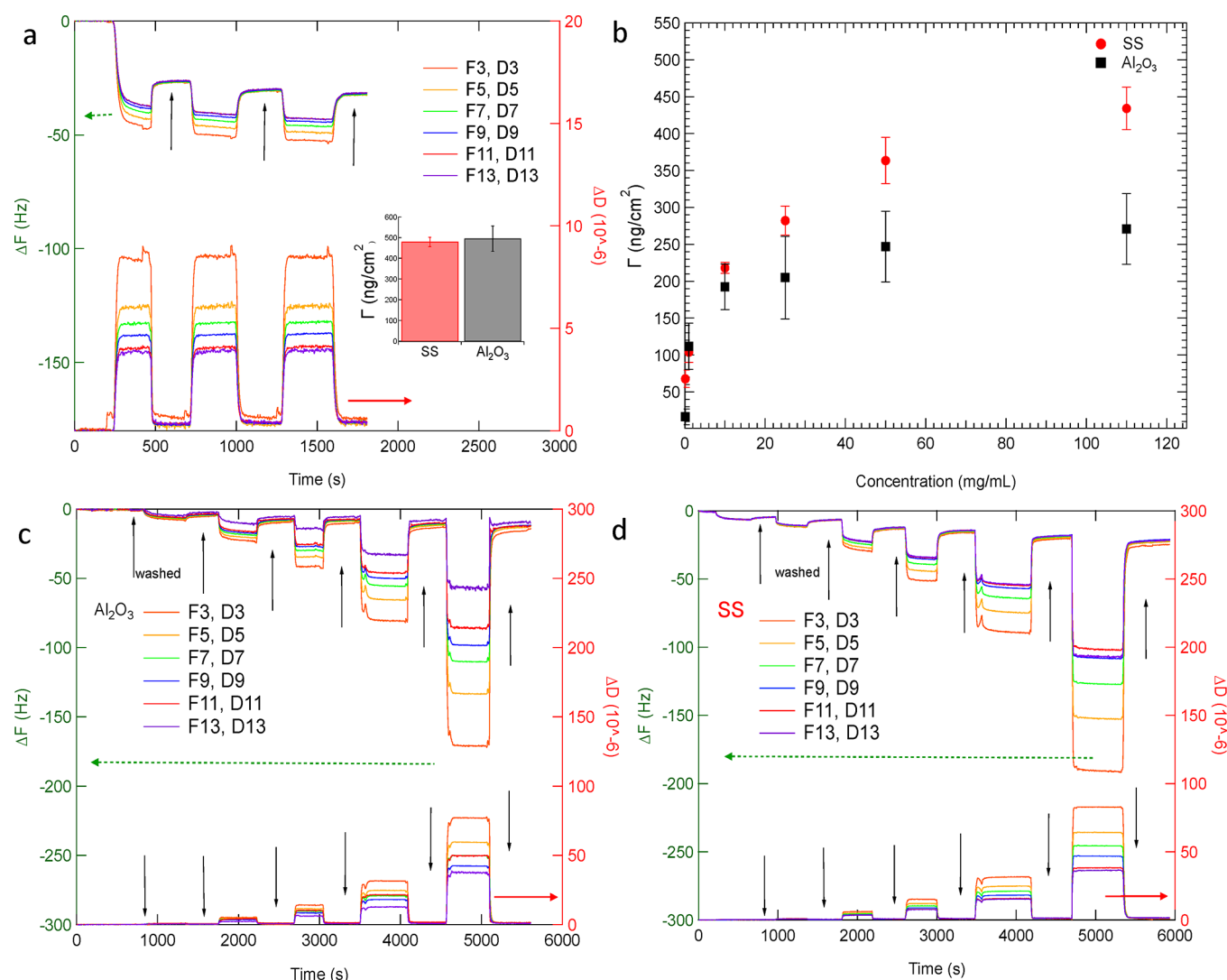


Figure 3. Summary of QCM-D experiments. (a) The adsorption of the buffer, i.e., no protein, on the stainless steel sensor. The inset of panel a compares the mass of the buffer solutes adsorbed and remaining during washing. Different harmonic overtones for frequency and dissipation (3, 5, 7, 9, 11, 13) of the quartz resonator are shown, which can be used to model the rheological properties of the contacting material at the solid–liquid interface. (b) The comparison of the irreversibly adsorbed mass of protein calculated by the Sauerbrey equation, $\Delta f = -\delta m \times \frac{2f_0^2}{A\sqrt{\rho_q\mu_q}}$, for the washed QCM-D sequential protein concentration adsorption measurements of the fusion protein on (c) SS and (d) Al_2O_3 sensors. The variables (Δf , δm , f_0^2 , A , ρ_q , μ_q) represent frequency change, mass change, resonant frequency, active crystal area, the density of quartz, and shear modulus of the quartz of a 5 MHz crystal. The green dashed arrow identifies the data for frequency changes, and the red arrows identify the data for changes in dissipation, ΔD . The black arrows point to the washing periods of the adsorption experiment. Mass uncertainties, ± 1 st. dev., reported for QCM-D are $n = 4$, calculated from the Sauerbrey equation.

mechanical or rheological properties of the adsorbed layers are sensitive to the separation between the harmonic overtones measured by the QCM-D sensor described previously. When the harmonics are separated, viscoelastic modeling can be used. Figure 3a shows the adsorption behavior of the buffer on the SS QCM-D sensor. Measuring adsorption using QCM-D compares the differential frequency and dissipation shifts with respect to a baseline solution. The buffer solution contains excipients (amino acids and salt) that can also adsorb onto the solid surface. Polysorbate 80 (PS80), a commonly used surfactant in pharmaceutical products, was intentionally omitted from the formulation to simplify the experimental adsorption conditions. The pumping studies contained PS80, but its presence showed minimal impact on preventing particle formation.⁴⁶ To test if the buffer interacts with the two surfaces

differently, we measured the adsorption behavior of the buffer with respect to deionized water. Interestingly, the harmonics for frequency and dissipation are separated during the adsorption period, indicating a viscoelastic layer of excipients, which then forms into a rigid layer with no dissipation after rinsing with deionized water. During the wash period, the harmonics collapse, and there is a decrease in magnitude. It is surprising that the magnitude does not return to zero. The behavior is repeated for two more cycles, showing an irreversible change. Similar results are observed for the Al_2O_3 sensor (ceramic mimic). We apply the Sauerbrey equation, $\Delta f \sim \delta m$, to quantify the mass of buffer adsorbed on the QCM-D sensor for the wash periods because the overtones shown for frequency show minimal separation and no dissipation.^{44,45} The inset in Figure 3a shows the apparent masses of buffer

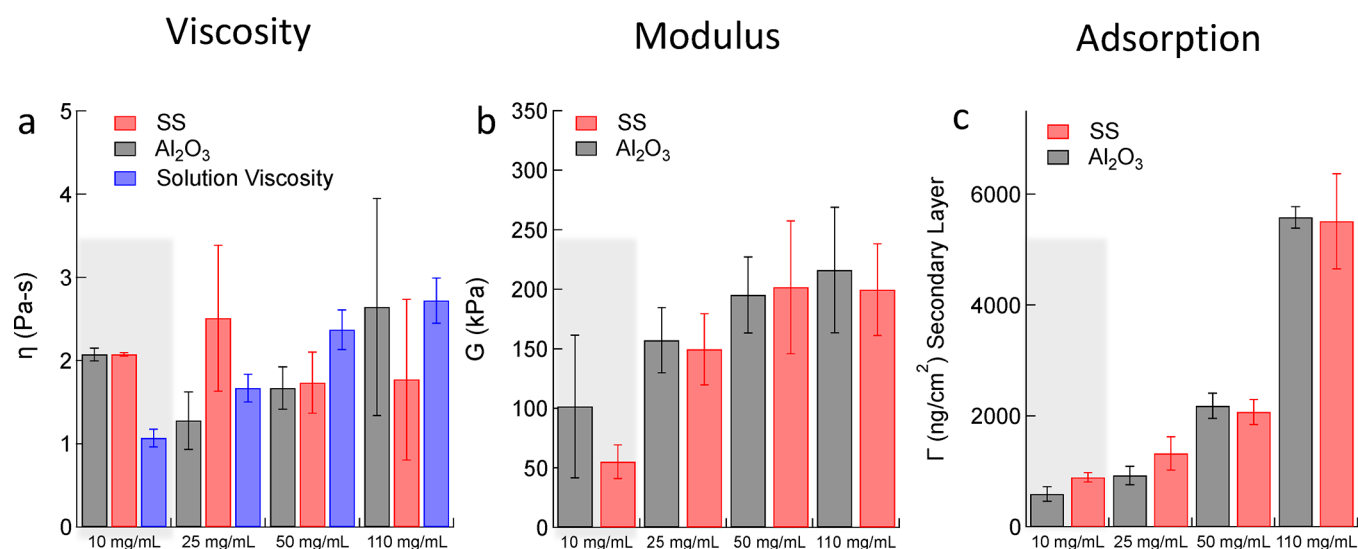


Figure 4. Physical properties. (a) Viscosity (η), (b) modulus (G), and (c) mass (Γ) of protein in the secondary layer for concentrations greater than 10 mg/mL. Uncertainties, ± 1 st. dev., reported for QCM-D are $n = 4$ calculated from the Voigt Model. Uncertainties, ± 1 st. dev., of physical properties reported are for $n = 4$.

species adsorbed onto the two different surfaces are similar. The thickness, t , of the adsorbed mass, assuming arginine hydrochloride (Arg-HCl) is the dominate adsorption species, is estimated to be $t_{\text{Arg HCl}} \sim 3$ nm, using a density ($\rho_{\text{Arg HCl}}$) equal to 1.459 g/mL. Arg-HCl has an end-to-end molecular distance of be around 0.7 nm. This suggests that the adsorbed film is a multilayer even if Arg-HCl is oriented perpendicular to the surface. For comparison, *n*-octadecyltrichlorosilane forms a self-assembled monolayer, where its molecular length, 2 to 3 nm, corresponds with its measured film thickness.⁴⁷ This multilayer and irreversible adsorption of buffer species is unexpected.

Figure 3c,d shows the adsorption behavior of the fusion protein in buffer solutions on SS and Al₂O₃, respectively. At low protein concentrations (0.1 mg/mL, 1.0 mg/mL), the shift in frequency, dissipation, and separation of harmonics is small compared to higher concentrations (>10 mg/mL) during the adsorption and wash periods. The results at low solution concentrations indicate a rigid film adsorbs on the surface. We call this the primary layer. At higher protein concentrations, there is a separation between the different harmonics for frequency and dissipation during the adsorption period. This indicates the adsorbed film on the QCM-D sensor is viscoelastic at high concentrations. Similar adsorption behavior is also observed for the fusion protein on the Al₂O₃ sensor. In contrast to the SS sensor, the magnitude of the frequency shift and dissipation of the wash periods are less compared to SS. To characterize the adsorption behavior, the simplest two-layer model is used to quantify the physical properties of the adsorbed films. These results are described with an analysis of the wash periods and adsorption periods.

Figure 3b compares the mass adsorbed for the wash periods for the two different surfaces. The Sauerbrey equation is applied to quantify this irreversibly adsorbed mass as was done for the adsorbed amount of buffer species because the overtones have minimal separation. Because the amount of adsorbed species from the buffer is comparable for both surfaces, it allows for a direct comparison of adsorption protein between the two surfaces. At low concentrations, both surfaces adsorb similar quantities of protein. At 10 mg/mL, the

thickness of the primary protein layer, t_{fusion} , is estimated to be between 1 to 2 nm, dependent on the assumed areal density.²⁵ Here, the Al₂O₃ sensor is fully saturated, whereas the SS sensor continues to adsorb more mass at the highest protein solution concentration tested. At 10 mg/mL, there is a transition point such that the harmonics show separation >2 Hz, and this is the last concentration where the two surfaces irreversibly adsorb the same amount of protein. The implications of the calculated film thicknesses for the primary protein layer with respect to SVP formation will be provided in the Discussion section.

We now turn to the analysis of adsorption at a higher protein concentration and reversible adsorption. We characterized the adsorbed film during the adsorption periods at concentrations of and greater than 10 mg/mL using a viscoelastic Voigt model because the QCM-D data show separated harmonics.⁴⁸ Further details of the modeling are described in the Supporting Information. From the raw data shown in Figure 3c,d, the adsorption behavior of this diffuse layer appears to be similar regardless of substrate. This impression is confirmed by the modeling quantified in Figure 4. Figure 4a–c compares the viscosity, modulus, and mass adsorbed of the second layer for SS and Al₂O₃. The film properties for the secondary layer at concentrations greater than 10 mg/mL show that its viscosity is comparable to the solution viscosity, and the moduli all have similar values of a soft gel as this is to be expected given that secondary layers primarily rely on protein–protein interactions. As the protein layers are farther from the solid surface, we expect the protein solutions in contact with the various surfaces to start behaving more similarly than that of layers in immediate proximity to SS or ceramic. As the protein concentration increases, the mass of the secondary layer increases, indicating this gel layer increases in thickness, assuming a constant film density, an assumption consistent with its nearly constant η and G (Figure 4a,b). For SS, this secondary layer is the softest, lowest modulus (G) at the transition concentration of 10 mg/mL. It does not appear that the secondary layer is predictive of the level of SVP formation. Given the complexity of possible SVP formation mechanisms, understanding these details will be important for

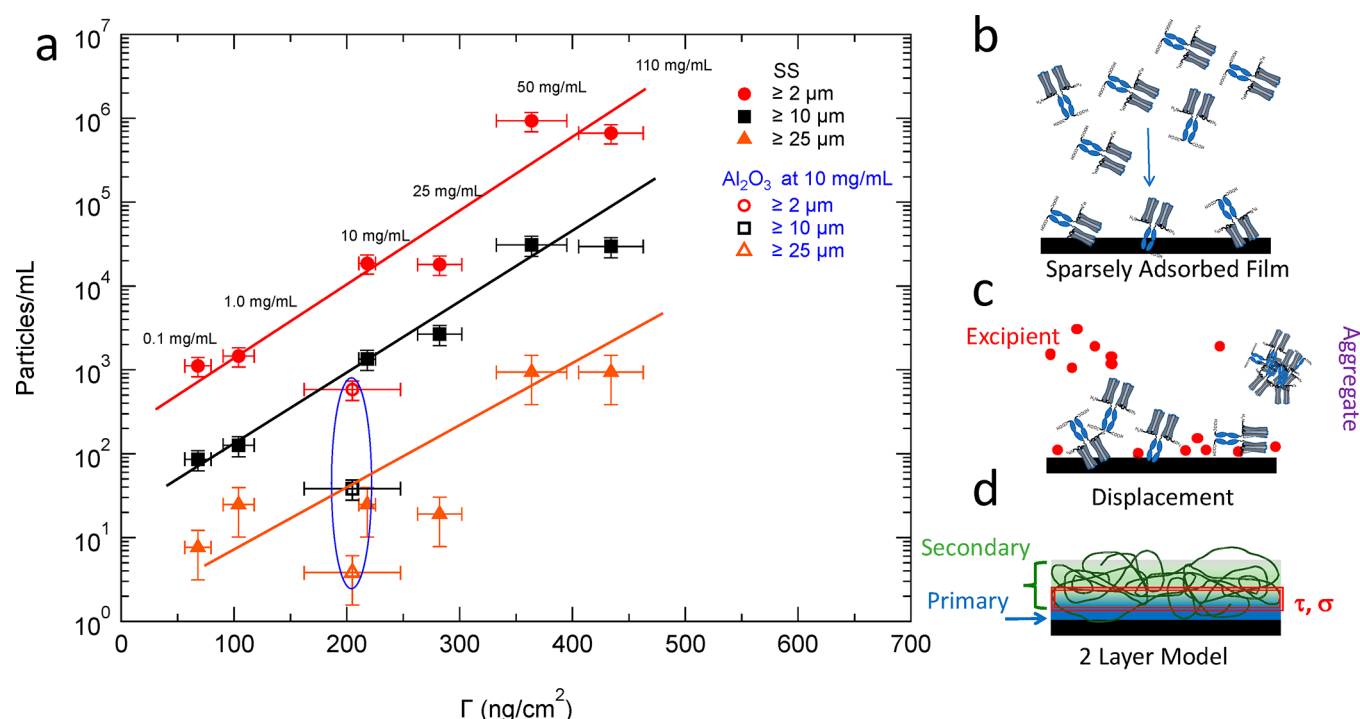


Figure 5. (a) The mass of the primary layer for the wash periods plotted measured by QCM-D adsorption experiments against SVP counts generated from the piston pumping experiments. For comparison, the mass adsorbed versus SVP counts for Al_2O_3 is shown for 10 mg/mL and circled. The lines are guided to the eye for clarity for data relevant to the SS surface. (b) The schematic illustrates the surface constitution for adsorption after washing. (c) Proposed mechanism for SVP formation, where ionic species displaces adsorbed protein similar to the Vroman effect. (d) A representation of the two-layer model suggested by QCM-D and NR data. Of interest is the interactions between the irreversibly rigid primary layer and the soft reversibly adsorbed secondary gel layer. The black layer represents the surface. The top layer, blue, represents the primary layer. Mass uncertainties, ± 1 st. dev., reported for QCM-D are $n = 4$ calculated from the Sauerbrey equation.

further study using other analytical methods such as neutron reflectivity.

DISCUSSION

Identifying the mechanism and cause for SVP formation is challenging because a number of factors, for example, interfaces, temperature, agitation,⁴⁹ light, impurities, etc.,^{7,30} could be responsible. It is further generally hypothesized that misfolded/partially unfolded protein molecules could have exposed, normally buried, hydrophobic residues, and/or hydrophobic molecule patches. These motifs can be highly susceptible to irreversible protein aggregation.^{50–52} Pinpointing the factors that cause SVP formation and its molecular origin are important to develop control strategies, including molecular design strategies, to specifically mitigate SVP formation. This study shows the SS interface as the root cause of increased SVP formation for this fusion protein. From the results, we develop a hypothesis that increased mass adsorption induces protein aggregates, which then nucleate. This ultimately results in a significant increase of SVPs in solution.

The surface binding affinity of the fusion protein on SS measured by QCM-D is stronger than that of Al_2O_3 because SS continues to adsorb more protein at higher concentrations irreversibly. The capacity for SS to adsorb more protein appears to be driven by attractive electrostatic interactions. The protein's glycosylated form used here has a measured pI between pH = 7.1 to 8.6, giving it a positive charge in the buffer of pH 6–7. Surface charge measurements, relevant to irreversible adsorption of the primary layer, show the SS

interface is negatively charged at this pH, whereas the Al_2O_3 surface charge is neutral or slightly positive.^{53,54} The positive charge of the fusion protein at this pH suggests that attractive electrostatic forces influence the adsorption mechanism onto SS promoting a higher amount of protein to adsorb onto the surface. A stronger binding affinity on the SS surface suggests that the more mass adsorbed onto the surface from the wash periods leads to SVP generation shown in Figure 5a compared to Al_2O_3 . To allow for an increase in adsorption suggests that the structural or orientation of the adsorbed protein has changed from its native state. A consequence of these rearrangements is not only increased mass adsorption but also the possibility that the non-native conformational form of the adsorbed protein could serve as nucleating sites for SVP. The relationship between an increase in protein mass and SVP generation has previously been observed with the NIST mAb.²⁵ At the air–water interface, the increased surface excess of protein has been shown to correlate with an increase in protein aggregation or a decrease in soluble mAb concentration.²⁴ The SS surface has been found to induce protein aggregation, but the debate on how it influences this process is still ongoing.^{22,28,29}

Analysis of the adsorbed layer of protein in dilute concentrations raises some interesting observations about the nature of the film for the wash periods. Regardless of surface orientation, the estimated film thickness corresponds to a sparsely adsorbed surface layer that is approximately a third of the total surface area (Figure 5b). Therefore, the “saturated” amount (200 ng/mL) on Al_2O_3 is quite dilute, which would allow the protein to spread out and change conformation. On

SS, these conformational changes make the adsorbed proteins more susceptible to aggregation as more mass is adsorbed onto the surface. Neutron reflectivity (NR) data of fusion protein adsorbed onto SS demonstrates that adsorbed protein density is highest close to the stainless surface and decreases with distance from the surface (Supporting Information). The overall thickness of the measured primary layer is approximately 11 nm, as determined by the modeled data. These observations are consistent with partial unfolding but do not unambiguously demonstrate that partial unfolding has occurred. Additional modeling of the neutron reflectivity data using molecular dynamics simulations could aid in the structural interpretation of the neutron reflectivity data, but such work is beyond the intended scope of this manuscript, and such capabilities will be explored in the future. An increase in protein mass on SS compared to Al_2O_3 results in the film that either becomes more dense or thicker. From these scenarios, protein destabilization can cause irreversible aggregation by an increase in protein–protein contacts compared to the bulk solution or changes in local protein environment.^{51,52} Protein destabilization could happen in the adsorbed layer for concentrated protein environments or adsorption periods, as well.

During the wash periods, the metal interface adsorbed with protein is exposed to dilute protein environments. Interestingly, QCM-D shows that ionic species in the buffer solution adsorb strongly to metal interfaces. The viscosities of deionized water and the buffer are not different from each other. The protein adsorbed in the washed regions is significantly less for both surfaces compared to the mass adsorbed from the buffer excipient layer. Currently, it is unclear whether or how much this buffer excipient layer is displaced by protein adsorption. If the strength of the interaction is stronger for the ionic species compared to the protein, these species could displace presumably denatured proteins to resuspend in solution. Studies of competitive adsorption of multicomponent mixtures of protein demonstrate the Vroman effect, where one protein, A, can displace another protein, B^{55,56} (Figure 5b).

Using the experimental adsorption protocol, exposure, and wash, we could determine the existence and properties of the adsorbed secondary layer adjacent to the denser primary layer. The physical properties of the secondary layer show little difference between SS and Al_2O_3 surfaces, but it is worth investigating the two-layer film as a possible source for SVP generation. At 10 MHz, QCM-D measures a soft gel interacting with the rigidly adsorbed primary layer. During pumping, this interface experiences a range of shear rates, giving rise to interfacial deformation and stresses in the primary and secondary layer, which could be consequential toward the formation of SVPs (Figure 5d). Examples have been reported that it is the interaction at interfaces that first induce a structural instability for the protein, and it is conjectured (without evidence) that the added stress from shear forces exerted on the molecules exacerbate the molecular deformation.^{23,28,57} At this point, the relatively unstable proteins are concentrated at or near the surface. These proteins can now readily interact to form aggregates and particles similar to the suggested mechanism of proteins in the rigid film. However, there are limited methods that are well suited to characterize the dynamics of this interstitial layer. From the data in Figure 4, the physical properties of the secondary layer are not different between the two surfaces. It is difficult to measure the rheological properties at the solid–

liquid interface. Particle tracking microrheology using particle tracking could be a suitable method to measure these properties.⁵⁸ The transition from concentrated to dilute protein environments is another pathway that could promote SVP formation. We have speculated several pathways for surface-mediated SVP formation from the QCM-D data, but the technique is limited.

QCM-D measurements do not provide molecular evidence at the metal protein interface or desorbing species. Other approaches, such as molecular spectroscopy⁵⁹ or scattering techniques,^{7,25} would complement the characterization of the physical and rheological properties of the thin film. Also, it would clarify the adsorption process for these formulation mixtures and the interplay between buffer excipients, proteins, and surface adsorption. There are minimal studies investigating the adsorption behavior process of a therapeutic protein in a similar way, to which equipment is used in the manufacturing process.²⁵ Exposing surfaces exposed to dilute and concentrated protein environments, monitoring adsorption/desorption processes, and characterizing the adsorbed film's physical properties have provided a platform to study the further interface-mediated (solid–liquid) SVP formation.

■ CONCLUSIONS

In this work, we pinpoint a surface-mediated cause, the SS metal interface, for an increase in SVP for a fusion protein. Stainless steel interfaces have been shown to mediate protein aggregation, but this work focuses on a novel fusion protein that demonstrated extreme particle formation upon passage through a stainless steel piston pump but not upon passage through a ceramic piston pump. Using QCM-D to quantify adsorption behavior, the results show there are differences between SS and Al_2O_3 (ceramic mimic). This raises the possibility that there are measurable attributes, in this case, mass adsorbed, that can be tied to predicting which surfaces will give acceptable levels of SVP for product quality and regulatory reporting. Understanding which surfaces are suitable for manufacturing reduces time, money, and unforeseen complications during production.

This study, coupled with previous studies, raises fundamental questions about adsorption behavior at the metal interface. Further study is needed in several areas to quantify better how surface-mediated interactions drive SVP formation. For example, systematic studies on the different surface properties such as charge, roughness, or chemical composition would then open a pathway for material design to prevent SVP formation. Molecular interrogation of adsorbed species of multicomponent solutions would provide direct evidence of protein surface interactions or structural changes for adsorbed proteins.⁵⁹ Novel measurement techniques that measure bulk and surface concentrations simultaneously would provide a better description of the adsorption/desorption process. Connecting critical quality attributes in therapeutic drug development such as SVP formation with fundamental surface studies could reduce unexpected challenges during product development.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.molpharmaceut.9b01000>.

Formulation characterization for different lots; particle analysis of subvisible particles (SVP); scanning electron microscopy elemental analysis and Fourier transform infrared spectroscopy; QCM-D modeling; example of quartz crystal microbalance with dissipation monitoring (QCM-D) fitting parameters of the two-layer model for different concentrations; the structure of the fusion protein adsorbed to stainless steel determined from modeled neutron reflectivity data (PDF)

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Notes

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